

YANAN HE

📍 New Haven, CT 🌐 <https://echoo113.github.io>

*I study why AI **works** when it works, why it **fails** when it doesn't, and how understanding those differences can help us **build better models**.*

EDUCATION

Yale University , New Haven, CT	Aug 2026 – May 2028
M.S. in Computer Science (<i>Fully Funded</i>)	
Purdue University , West Lafayette, IN	Aug 2023 – May 2026
B.S. in Computer Science & Mathematics-Statistics	GPA: 3.96/4.0 <i>Graduated with Distinction</i>

RESEARCH EXPERIENCE

NLP Lab, Yale University	<i>New Haven, CT</i>
Research Assistant Advisor: Arman Cohan	Apr 2026 – Present

- **Model Behavior, Interpretability & Post-Training** – investigating the mechanisms underlying **subliminal learning** and using these insights to improve how language models learn during post-training.
- **LLM-as-a-Judge for Deep Research Agents** – developed REFLECT, a fine-grained benchmark for evaluating **judge reliability** across agent reasoning, tool use, and final reports [1]

Graph Lab, Stony Brook University	<i>Stony Brook, NY</i>
Research Assistant Advisor: Tengfei Ma	May 2025 – Apr 2026

- **Time-Series Modeling** – mitigated **latent prediction collapse** in JEPA with structured codebooks [2]; detected **causal residual shifts** via multi-scale structural consistency [3].
- **LLM Reasoning in Geometric Space** – guided multi-step reasoning with **hyperbolic latent signals** [4].

SELECTED PUBLICATIONS

1. Time to REFLECT: Can We Trust LLM Judges for Evidence-based Research Agents? * Equal contribution.
Leyao Wang*, **Yanan He***, Peng Chen, Asaf Yehudai, Yixin Liu, Rex Ying, Michal Shmueli-Scheuer, Arman Cohan
arXiv:2605.19196.
2. SC-JEPA: Stabilizing Latent Predictive Learning for Time-Series Anomaly Prediction
Yanan He, Yunshi Wen, Xin Wang, Tengfei Ma
SDM 2026; Forecast@ICML26.
3. CAAD: Causality-Aware Multivariate Time Series Anomaly Detection via Multi-Scale Alignment and Structural Causal Consistency
Xin Wang, Yunshi Wen, **Yanan He**, Haotian Xu, Youlan Zhao, Michel Ferreira Cardia Haddad, Tengfei Ma
SIGKDD 2026 Research Track.
4. HyperGuide: Hyperbolic Guidance for Efficient Multi-Step Reasoning in Large Language Models
Yuyu Liu, Haotian Xu, **Yanan He**, Sarang Rajendra Patil, Mengjia Xu, Tengfei Ma
arXiv:2605.24140.

SKILLS

- **Programming Languages:** Python, Java, C/C++, Assembly, R, SQL, MATLAB, JavaScript, HTML/CSS
- **Frameworks & Tools:** PyTorch, Hugging Face, NumPy, Pandas, scikit-learn
- **Languages:** English (Professional), Chinese (Native)